Feature Selection for Data-Driven Solar Photovoltaic Maximum Power Point Tracking

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***Abstract*—Reliable Maximum Power Point Tracking (MPPT) is essential for optimizing solar photovoltaic (PV) plant efficiency under dynamic environmental conditions. While modern data-driven MPPT implementations rely on intelligent machine learning and deep learning models to predict the optimal reference voltage, existing designs frequently utilize plane-of-array irradiance and module temperature as sole predictors without systematically evaluating secondary meteorological variables. This study establishes a systematic feature selection framework using 1,196,831 real-world operational samples collected over 4 years across 9 distinct PV modules and 7 technologies. Linear and non-linear dependencies are quantified utilizing univariate metrics alongside multivariate structures, including F-regression, Mutual Information, Lasso regression, Recursive Feature Elimination (RFE), and Permutation Importance. The empirical results demonstrate that a core subset of 3 features (irradiance, module temperature, and ambient temperature) drives tracking performance with high stability across all evaluated technologies. Secondary atmospheric parameters, including wind speed, wind direction components, relative humidity, and absolute pressure, exhibit negligible relevance and can be omitted to minimize costs and computational load for deployment.**

***Index Terms*—Maximum power point tracking, photovoltaic systems, feature selection, machine learning, real-world data.**

I. INTRODUCTION

The efficiency of solar photovoltaic (PV) systems relies on operating at the maximum power operating point, which is highly sensitive to variations in environmental conditions, mainly irradiance and temperature [1]–[3]. Conventional Maximum Power Point Tracking (MPPT) methods oscillate, converge slowly, and tend to fail under rapid weather variations [4]–[6], reducing efficiency by 5 to 30% [7]. State-of-the-art MPPT implementations leverage diverse deep learning architectures, particularly Multilayer Perceptrons [8]–[10] and recurrent neural networks [7], [11]. These methods are highly dependent on the quality of data used for training [6], [12]. Furthermore, current literature relies entirely on irradiance and module temperature as input features for data-driven models, ignoring secondary environmental variables that affect solar yield. This results in models with a surplus or deficiency of inputs, increasing costs and decreasing the algorithm accuracy. We address this limitation by providing a feature selection framework to determine the relevant features for data-driven MPPT, reducing the sensor set to the most crucial. Dropping the least relevant sensors reduces computational load and system complexity, enabling low-cost deployment of MPPT [13].

We utilized univariate and multivariate feature selection methods on an extensive real-world dataset that combines electrical and environmental variables, comprising 1,196,831 operational samples across nine PV modules and seven different technologies.

This paper is structured as follows: Section II presents the real-world data acquisition and preprocessing pipeline. Section III explains the methodology for feature selection, including target normalization and feature engineering, the analysis of feature dependency, and univariate and multivariate feature selection methods. Section IV analyzes and compares the results of these methods. Finally, Section V concludes the paper, summarizing the key findings.

II. REAL-WORLD DATA ACQUISITION AND PREPROCESSING

We collected experimental real-world data at the Outdoor Photovoltaic Research Laboratory of the Pontificia Universidad Católica del Perú (PUCP), located in the coastal area of Lima, Peru (12°04′ S, 77°04′ W). The system comprises an I–V curve tracer synchronized with a comprehensive monitoring system that records plane-of-array irradiance (G_{poa}), spectral distribution, module temperature (T_m), and relevant meteorological variables [14]. This configuration enables simultaneous electrical and environmental characterization of the PV modules under outdoor operating conditions in the distinctive climate of the city, an arid hot desert climate with high annual relative humidity levels [14], [15].

Monofacial PV modules were mounted at a 20° north-facing tilt angle, as illustrated in Fig. 1. Bifacial PV modules were installed in a 15° north-facing tilted configuration, as described in [16].

Irradiance at the plane of the array (G_{poa}) was measured using EKO MS-80 pyranometers. Ambient meteorological parameters, including ambient temperature (T_a), relative humidity (RH), wind speed (v_w), wind direction (θ_w), air density (ρ_{air}), and absolute pressure (P_{abs}), were recorded using a LUFFT WS500-UMB smart weather station, while module temperature (T_m) was monitored using thermocouples placed

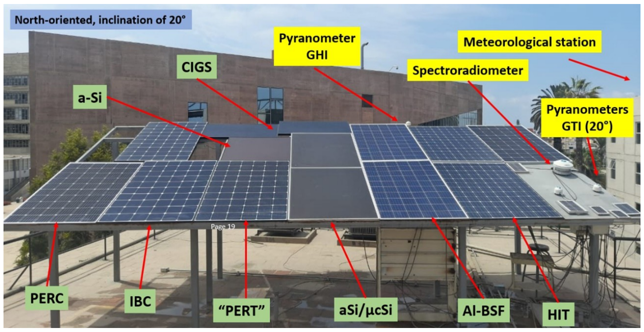

Fig. 1. Outdoor Photovoltaic Research Laboratory at PUCP

TABLE II
PV MODULE SPECIFICATIONS AND DATASET DISTRIBUTION

Module Name	Rated Power (W)	Technology	Sample Ratio (%)
QPEAK315	315	PERC	13.64%
LG370Q1C	370	Mono-Si	13.56%
LG345N1C	345	Mono-Si	13.36%
CS6K270P	270	Poly-Si	13.33%
VBHN330	330	HIT	12.84%
CDF1150A1	115	CIGS	11.94%
NAF128GK	128	a-Si/μc-Si tandem	11.16%
CS3K320MB	320	PERC	7.18%
CIEH5C1325	325	HJT	3.00%

on the side and center of the panels. All PV modules and sensors were cleaned weekly to minimize the effects of dust accumulation.

The maximum power point (MPP) was identified by computing the instantaneous electrical power as the product of current and voltage and locating its maximum value along the curve. The corresponding voltage and current values were then assigned as the MPP operating point. Environmental parameters were extracted directly from the corresponding measurement instrument.

We filtered the data based on the instruments' measurement limits and physical constraints. After the initial filter stage, interquartile range filtering was applied to account for extreme outliers. Rows with filtered values were discarded, given the large quantity of data available. Additional filters were implemented to account for sensor errors, partial shading, and significant variations of irradiance in a single measurement. Table I describes these filters, where STC refers to Standard Test Conditions (1000 W/m^2, 25°C, AM1.5).

TABLE I
DESCRIPTION OF DATASET FILTERS

Variable	Filter	Description
Irradiance	Range	Recorded irradiance matches computed spectral irradiance.
Irradiance	Variation	Relative difference between irradiance measured at the start and end of the curve is less than 4%.
Temperature	Range	Difference between lateral and central temperature is less than 25°C.
V_{MPP}	Range	Must be [20% – 100%] of V_{OC} at STC.

The rated power and technology of the nine distinct PV modules used for this study, as well as the distribution of the 1,196,831 samples spanning 4 years of operational data, are shown in Table II.

Data from 2022 to 2024, comprising 925,959 samples, were utilized for the training and validation set, while data from 2025, comprising 270,872 samples, were withheld as a hold-out test set for evaluation.

III. FORMULATION OF FEATURE SELECTION

To systematically determine the minimal sensor set required, the feature selection framework evaluates both individual feature contributions and joint multivariate interactions. In solar PV modeling, physical relationships are a mix of linear (e.g., module temperature T_m scaling with ambient temperature T_a) and non-linear (e.g., exponential I–V diode characteristics). Therefore, our framework integrates both linear and non-linear univariate and multivariate models to capture the full spectrum of predictor-target dependencies across diverse PV panel technologies.

A. Target and Feature Engineering

To enable unified feature selection across diverse panel capacities and technologies, the electrical target variable is isolated as a dimensionless normalized voltage ratio, defined by Eq. (1).

$$\gamma_v = \frac{V_{MPP}}{V_{OC,STC}} \tag{1}$$

where V_{MPP} is the measured instantaneous maximum power-point voltage and $V_{OC,STC}$ is the specified open-circuit voltage at STC.

Raw wind direction vectors (θ_w), specified in angular degrees, present mathematical discontinuities at the $0°/360°$ boundary. To resolve this boundary constraint, the raw data were mapped into continuous cyclical spatial components using sine and cosine trigonometric transformations ($\sin(\theta_w)$ and $\cos(\theta_w)$).

B. Feature Dependency

The geographical location of a solar plant affects the environmental conditions the panels are exposed to. While most studies rely on irradiance and temperature as the most important features for MPPT [7], [9], [11], [12], there are other factors that affect solar yield. For instance, module temperature is directly affected by irradiance, air temperature, humidity, and wind speed [17], indirectly contributing to MPP shifting, since PV voltage is most affected by temperature. Module temperature is linearly correlated with air temperature, increasing 0.85°C for every 1°C increase on the latter [18]. According to the second law of thermodynamics, heat flows from hotter to colder regions; therefore, a colder location leads to decreasing cell temperature. Wind speed and direction also play a part in thermal dissipation [19], [20], further decreasing module temperature in high-wind-speed locations [21]. On the

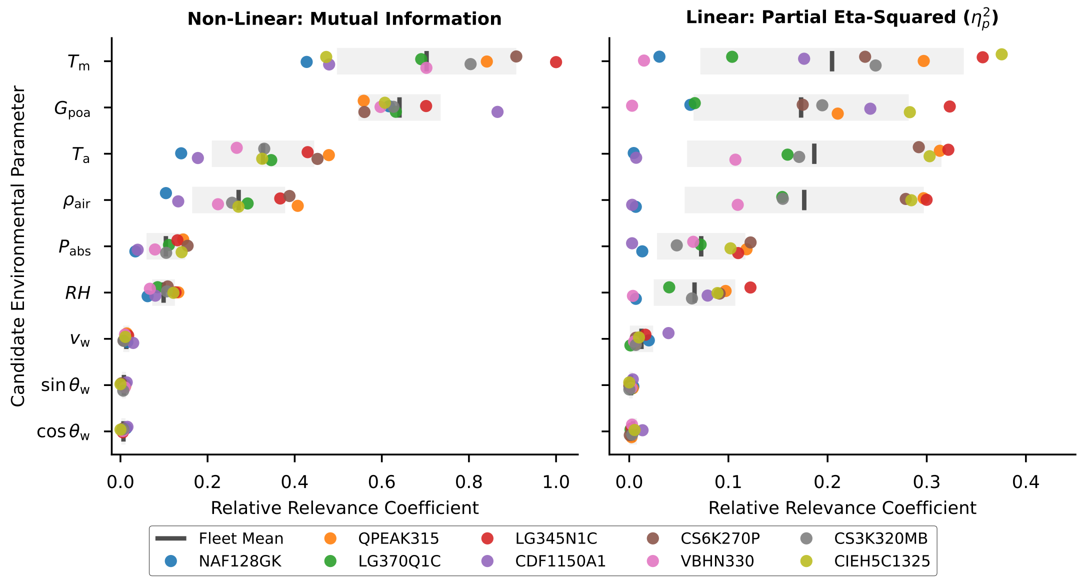

Fig. 2. Univariate Feature Importance: Relative Relevance Coefficient from MI and η_p^2

other hand, it has been noted by [22] that relative humidity has a negligible effect on module temperature. However, it can affect irradiance dispersion, affecting the amount finally reaching the panel.

C. Univariate Methods

Initial feature screening evaluates both linear and non-linear dependencies across the heterogeneous PV modules. Non-linear relationships are quantified using Mutual Information Regression (MIR), which estimates the entropy between environmental inputs and the target γ_v. MIR is a non-negative measurement of the dependency between two variables; a value of zero indicates that both variables are independent, while larger values mean higher dependency [23]. Because raw MIR scores lack fixed upper bounds, the outputs were globally normalized to a [0,1] scale relative to the absolute maximum score across all modules, enabling direct cross-technology comparison.

To evaluate linear dependencies without introducing sample-size bias, feature effect sizes are quantified. Standard univariate linear filters output raw F-statistics; however, the absolute magnitude of an F-score scales monotonically with the number of samples (N). Because the dataset contains varying operational sample counts per PV module, comparing raw F-scores directly distorts the analysis. To resolve this, raw F-statistics are analytically transformed into sample-invariant partial eta-squared (η_p^2) effect sizes (Eq. (2)).

$$\eta_p^2 = \frac{F \cdot p}{F \cdot p + (N - p - 1)} \tag{2}$$

where $p = 1$ for univariate testing and N is the sample count of the target module. In this configuration, η_p^2 maps the percentage of target variance explained by each individual environmental parameter onto a stable baseline.

The use of both methods is justified by the property of MIR to measure any kind of relationship between variables [24], while η_p^2 only identifies linear relationships.

D. Multivariate Methods

To address the effect of multiple variables, Lasso Regression applies L_1 regularization to penalize absolute feature weights, driving irrelevant feature coefficients to zero to produce a sparse model [25]. To account for the chronological nature of the dataset, the regularization penalty was optimized using 5-fold Time Series Cross-Validation (TSCV), preventing data leakage. The resulting coefficients were normalized to a $[0, 1]$ scale per PV module to track structural elimination patterns across the modules.

Recursive Feature Elimination (RFE) determines the optimal feature subset by iteratively training a regressor and pruning the least important input variable, one at a time. The subset quality was evaluated using the Root Mean Square Error (RMSE) of the normalized voltage target (γ_v). TSCV was again utilized, defining the optimal feature subset as the combination that minimized the validation RMSE before it plateaued.

Permutation Importance isolates the contribution of each feature by randomly shuffling a single feature's data and observing the degradation in the model's score [26]. A dedicated Random Forest Regressor was trained on the 2022–2024 data for each PV module. Permutation testing was then applied to the 2025 hold-out test set. The resulting performance drop, measured in Mean Squared Error (MSE), was normalized per module, quantifying the reliance of the model on each feature when processing unseen data.

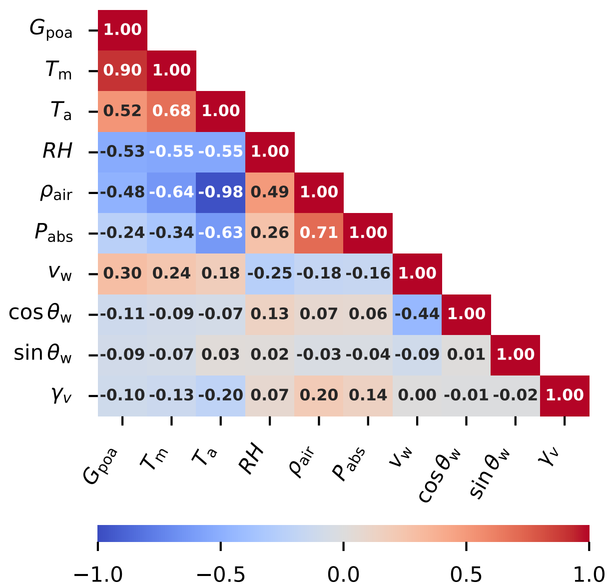

Fig. 3. Environmental Features and Target Correlation Matrix

IV. Results and Discussion

Fig. 3 presents the correlation matrix of environmental features and the target, showing the independent linear relationship between them and collinearity between features. Collinearity indicates that the information is redundant between two features. G_{poa} and T_m are highly correlated, an expected behavior given the known connection presented in the previous section and physical intuition; higher irradiance naturally heats up the panel. Similarly, T_a has a positive correlation with the aforementioned features. However, air temperature has a significant negative correlation with ρ_{air}, RH and P_{abs}; it is affected by the particular characteristics of the location. None of the features has a strong correlation with the target γ_v, indicating a non-linear or multivariate relationship with the features.

The results of the univariate feature selection methods are shown in Fig. 2. The colored points indicate the individual relevance coefficient for each panel, the gray vertical line indicates the mean value across all panels, and the light gray box depicts the standard deviation. P_{abs}, RH, v_w, $\sin\theta_w$ and $\cos\theta_w$ have low relative relevance in both linear and non-linear methods for all panels, with small variance between panels. These features are not independently relevant for predicting γ_v, and this behavior is consistent across all panel technologies. For T_m, G_{poa}, T_a and ρ_{air}, the univariate results are inconclusive, as shown by the high standard deviation for both methods. For some panels, such as LG345N1C (Mono-Si), these features consistently show high relevance, whereas for NAF128GK (a-Si/μc-Si), relevance is very low.

Lasso regression elimination of irrelevant features is depicted in Fig. 4. This method predominantly chose T_a as the most relevant multivariate predictor, while greatly reducing the weight of the rest of the features. The blank tiles represent features with zero weight, which were effectively eliminated by this method. ρ_{air} and $\sin\theta_w$ were consistently eliminated in all cases, while $\cos\theta_w$ and v_w were eliminated in eight out of nine cases. v_w is particularly relevant as a multivariate feature for the VBHN330 (HIT) panel, in contrast to the low relevance found with the univariate methods. T_m was eliminated in two cases, while maintaining a relatively low relevance among most panels, except for CS3K320MB and CIEH5C1325. Since T_m and T_a were demonstrated to be highly correlated, it was expected that Lasso regression would preserve one over the other. This behavior is driven by the properties of the L_1 regularization penalty, which randomly selects a single representative feature from a group of highly collinear variables.

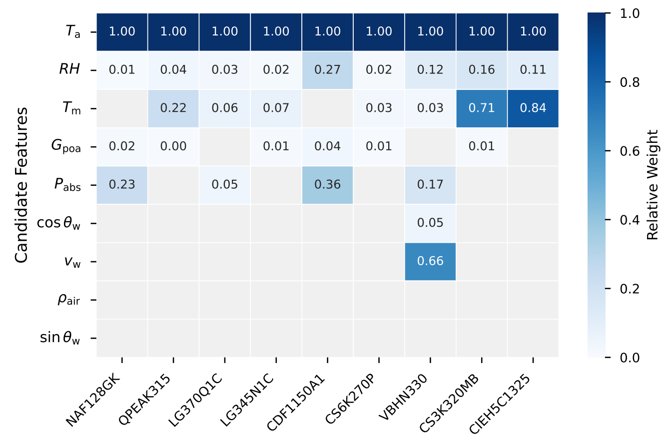

Fig. 4. Lasso Regression: Normalized feature weights per panel

Fig. 5 shows the selection consensus for all panels. G_{poa}, T_m and T_a were selected in all nine cases, reinforcing their relevance consistently throughout all panels. The absolute air pressure and wind variables were eliminated in all optimal subsets of features.

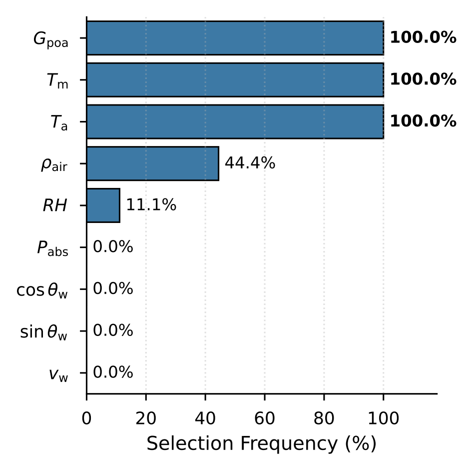

Fig. 5. RFE: Selection frequency across 9 modules

The RMSE over the number of features k presents an elbow at $k = 3$ (Fig. 6); additional features have little effect on the error. As depicted in Fig. 5, these 3 most relevant features are consistently G_{poa}, T_m and T_a across all panels.

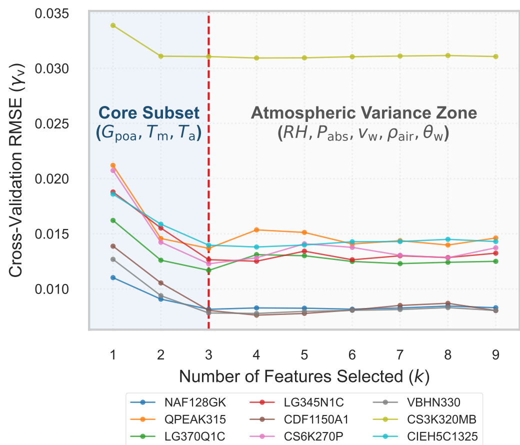

Fig. 6. RFE: RMSE decrease over number of features selected

Fig. 7 depicts the decrease in performance per feature, with each panel represented by a colored dot. Randomly shuffling T_a, $\cos\theta_w$, RH, $\sin\theta_w$, ρ_{air}, P_{abs} or v_w independently has very little effect on the performance of the model; this behavior is consistent across most panels. However, G_{poa} has a major effect on the performance of the model; without this feature, it loses most of its predicting capability. The relevance of module temperature varies widely across all panels, from close to no importance (CDF1150A1) to fundamental (LG345N1C); despite that, it remains the other most important feature on average for this method.

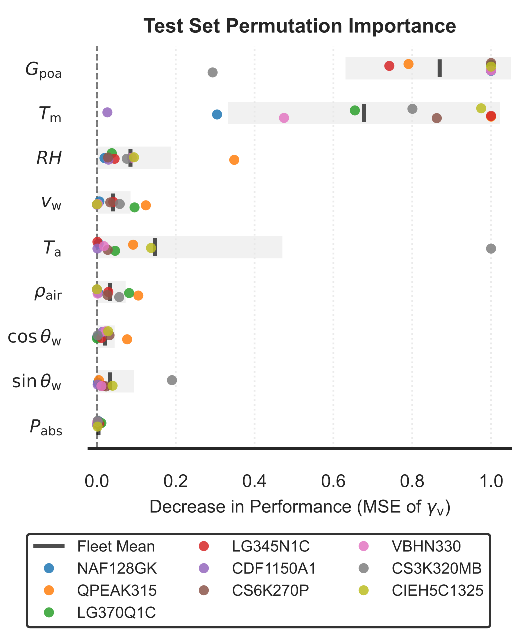

Fig. 7. Permutation Importance: Decrease in Performance

V. Conclusion

This paper presented a feature selection framework to establish the optimal subset of features for data-driven MPPT. Utilizing an extensive dataset with different PV technologies demonstrated that while distinct PV technologies exhibit localized variations in predictor-target sensitivity, the global feature importance tendencies remain highly stable. The empirical consensus across all univariate and multivariate selection methods identifies a 3-feature core subset, comprising plane-of-array irradiance (G_{poa}), module temperature (T_{m}), and ambient temperature (T_{a}) as the optimal configuration for tracking prediction. Conversely, secondary microclimatic variables, particularly wind-derived components ($\sin\theta_{\text{w}}$, $\cos\theta_{\text{w}}$, and v_{w}), absolute pressure (P_{abs}), and relative humidity (RH), show negligible standalone or multivariate relevance. Eliminating these secondary parameters allows for a significant reduction in sensor overhead and computational load, directly enabling the deployment of data-driven MPPT algorithms on low-cost setups. These findings are inherently connected to the local coastal desert climate of Lima, characterized by low average wind speeds and high relative humidity. In different climates with higher wind speeds and varying humidity, it is expected to observe a higher relevance of these features. However, irradiance, module temperature and ambient temperature are likely to remain the main drivers of maximum power point variation. Future work will focus on validating this feature selection hierarchy across diverse geographic microclimates to address for these differences.

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